Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/28/2004
PI Name:	Moorman, Stephen J Ph.D.		
Project Title:	The Effects Of Microgravity On Gene Expression In Live Zebrafish Embryos.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	08854-5635	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1591		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effects of microgravity on gene expression are poorly understood. Initial attempts have been made to investigate the effects of microgravity on cells in culture. It is unknown how those experiments will relate to potential effects on gene expression in vivo. We have begun to examine the effect of simulated-microgravity on gene expression in specific developing organ systems in vivo using transgenic zebrafish that express the green fluorescent protein gene (gfp) under the influence of different promoters. The use of gfp as a "reporter-gene" has two significant advantages; you can monitor gene expression directly in a live animal, and you can easily detect changes in morphology that might be due to changes in expression of other genes. This is the first time a whole animal in vivo model will be used and maintained alive during gene expression analysis for microgravity related research. This research not only addresses an important issue in space life sciences research, it clearly demonstrates the usefulness of zebrafish with gfp constructs as a model to		

	study the effects of microgravity on gene expression during development. Use transgenic zebrafish that express the gene for GFP under the influence of specific promoters to determine the effects of simulated-microgravity on gene expression in live zebrafish embryos. We have documented specific changes in gene expression that are related to changes in gravitational force. This suggests that gravitational force plays a fundamental role in regulating gene expression and might have been a driving force for vertebrate evolution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth's gravitational field has remained constant during all of evolutionary history. Therefore, it is possible that gravity might play some fundamental role in regulating gene expression during vertebrate development. If we persist in pursuing the idea of colonizing space and/or other planets, it is inevitable that a woman will become pregnant in space. We do not know whether a child conceived and/or born in space will develop normally. Our work on zebrafish suggests that there could be significant effects of microgravity on gene expression during development in vertebrates, including humans.
Task Progress:	In addition to documenting the effects of simulated-microgravity, we built a centrifuge to produce 2g and 3g for developing zebrafish embryos. This allows us to determine the effects across a continuum of changes in gravitational force. Progress: Beta-actin:GFP transgenic fish: We have demonstrated effects of changes in gravitational force on the expression of this cytoskeletal transgene in specific organ systems and have determined the developmental periods of maximum susceptibility for each system. In general, there is an up regulation in simulated-microgravity and a down regulation in hypergravity. The magnitude of these changes varies by organ system with the heart and notochord showing the most dramatic changes. HSP70:GFP transgenic fish: We have documented the effects of simulated-microgravity on this stress related transgene. In general, there is very little if any change in expression except in the skin, lens, and hypochord. Fli1:GFP transgenic fish: We have now demonstrated that both simulated-microgravity and hypergravity cause changes in expression of this blood vessel specific transgene. The changes involve an up regulation in simulated microgravity and a down regulation in hypergravity. In addition, the changes are more dramatic in blood vessels developing in the trunk than in those developing in the head.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Abstracts for Journals and Proceedings	Moorman, S.J.; Hoff, J.K.; Sokunbi, G. "Changes in gravitational force induce changes in gene expression in the developing zebrafish heart." Experimental Biology Meeting, Washington DC. FASEB Journal. , Apr-2004
Abstracts for Journals and Proceedings	Kwak, S.; Moorman, S.J; Riley, B. "Pax5 regulates neuroblasts production in the inner ear and is specifically required for vestibular function." International zebrafish meeting, Madison WI. Proceedings of International Zebrafish Development and Genetics Meeting. , Jul-2004
Abstracts for Journals and Proceedings	Cheng, K.C; Moorman, S.J. "A lifespan virtual atlas of the zebrafish as a starting point for systems morphogenetics." International zebrafish meeting, Madison WI. Proceedings of International Zebrafish Development and Genetics Meeting. , Jul-2004
Articles in Peer-reviewed Journals	Gillette-Ferguson, I.; Ferguson, D.G.; Poss, K.D.; Moorman, S.J. "Changes in gravitational force induce alterations in gene expression that can be monitored in the live, developing zebrafish heart." Advances in Space Research 32(8):1641-1646, Oct 2003. , Oct-2003